MagicCode Super-Resolution Software FAQ and Troubleshooting

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1. General

What does MagicCode Super-Resolution do?

It upscales images or GPU frames. v3.0.0 public SR modes are SR_SPEED_MODE_FOR_IMAGE, SR_BALANCED_MODE_FOR_IMAGE, SR_SPEED_MODE_FOR_GRAPHIC, and SR_BALANCED_MODE_FOR_GRAPHIC. GEN_SPEED_MODE_FOR_GRAPHIC and GEN_BALANCED_MODE_FOR_GRAPHIC add Vulkan frame generation where MAGIC_SR_HAS_FG is defined.

Does the software upload my images or videos?

No. Technical reporting does not upload image or video content. See the Privacy Policy and Data Reporting Notice (unchanged by this technical release).

What scale factors are supported?

Image: [1.0, 8.0]. Graphic: (1.0, 8.0]. Actual availability depends on backend and model. API support is not a claim that every scale is device-run.

2. Graphic FAQ

Is MC_Enable still available?

No. v2.2.0 removed MC_Enable and mc_enable.h. Use mc_nscaler_enable; graphic frames attach a temporal_frame_t on mc_nscaler_input_frame_t.frame.

If I omit masks, do I have a semantic transparency mask?

No. The library derives an approximate mask when inputs are missing. Explicit textures win per channel. FSR-family paths bind a zero transparency texture when omitted and do not generate semantic transparency from color.a.

Which backends and platforms support graphic?

Android Vulkan/OpenGLES, Apple Metal, Windows Vulkan / Direct3D 11 / OpenGL 4.3, Linux Vulkan (product libmagic_sr.a/.so). CPU graphic is unavailable. FG is a separate GEN path, Vulkan-only; Android Vulkan currently has no FG.

Is there a Linux build?

Yes. lib/linux/libmagic_sr.a and libmagic_sr.so provide x86_64 CPU image (AVX2/FMA) plus Vulkan Graphic (Vulkan Temporal)/TAAU and FG. Rebuild with tools/build_linux_lib.sh and link -lm -lpthread. CPU graphic is unavailable.

Does v3.0.0 change mc_nscaler_enable?

Yes. It takes mc_nscaler_input_frame_t *in_frame[2]. For SR pass { &in, NULL }. Setting in_frame[1] on a GEN handle interpolates t=0.5.

Which platforms support frame generation?

Vulkan only, and only when the library is built with MAGIC_SR_HAS_FG (Windows and Linux product libraries). Android Vulkan currently does not define it. D3D11 / Metal / GLES / OpenGL / CPU return -100050 / -102013.

Where does GEN_BALANCED load weights?

Set ctrl_param_t.model_path to the FG weight file. Empty path uses MAGIC_FG_WEIGHT_PATH, then ./magic_fg_core.bin, then model/magic_fg_core.bin. model_path is immutable; recreate the handle to change weights. The packaged file is model/magic_fg_core.bin (331,136 bytes). GEN_SPEED needs no weights.

- Vulkan Speed (Lite-TAAU) now lets an explicit strong reactive mask suppress the still-camera accumulation branch. Previously that branch ignored the caller reactive mask when the camera was still or per-pixel MV was below about 0.2 px, so pixels with ~0 MV but changing content (TV / video-wall / LED-screen textures) could keep ghosting for about 64–96 frames after a content cut. Strong reactive (≥ about 0.85, fully effective at ≥ 0.95) now weakens that accumulation. Omitting reactive or passing a weak value (< 0.85) is unchanged. Recommendation: mark screen-like objects whose MV is 0 but whose content changes (for example with CustomDepth stencil) as reactive at strength 1.0. Limit: Vulkan only in this release. The Metal / GLES / D3D11 Speed still-accumulation structure is the same idea but is not modified or verified here.

Graphic Super-Resolution

Graphic super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SR_SPEED_MODE_FOR_IMAGE) and processes the first frame. Pass mc_nscaler_input_frame_t *in2[2] = { &in, NULL } (set in2[1] only for GEN interpolate).

Modes

Value	Enumerator	Role

0	SR_SPEED_MODE_FOR_IMAGE	Image, throughput first
1	SR_BALANCED_MODE_FOR_IMAGE	Image, quality/cost trade-off
2	SR_SPEED_MODE_FOR_GRAPHIC	Graphic, throughput first
3	SR_BALANCED_MODE_FOR_GRAPHIC	Graphic, canonical FSR-family path
4	GEN_SPEED_MODE_FOR_GRAPHIC	FG-Fast (analytic, no weights). Vulkan + MAGIC_SR_HAS_FG. Combinable with SR bits; bare 4 equals SR_SPEED_MODE_FOR_IMAGE|GEN_SPEED.
8	GEN_BALANCED_MODE_FOR_GRAPHIC	FG-CoreV3. Same availability as Fast. Needs magic_fg_core.bin. Do not set both GEN bits.

- Graphic scaler_factor is (1, 8]; 1.0 itself is rejected. Image remains [1.0, 8.0].
- SR_SPEED_MODE_FOR_GRAPHIC on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family graphic path (same family as Balanced, not a separate customer API).
- SR_BALANCED_MODE_FOR_GRAPHIC uses the canonical FSR-family path on every GPU graphic backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run graphic modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan graphic requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.

- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both graphic modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

3. Initialization problems

- Model not found: SetModelPath/SetModelDir, packaging, permissions.
- Width/height out of range: 64..4032.
- Scale out of range: image [1,8], graphic (1,8].
- Backend unavailable: CPU graphic, or GPU graphic not in this build.
- Vulkan/D3D11 GPU create failed: missing gpu_context.device (and Vulkan physical_device).
- GEN not available (-100050 / -102013): non-Vulkan or build without MAGIC_SR_HAS_FG (including current Android Vulkan).
- FG weights (-100051 / -102014) or FG create (-100052 / -102015) on GEN_BALANCED.

4. Processing problems

- MC_ERROR_TEMPORAL_NULL_FRAME: mc_nscaler_input_frame_t.frame is NULL.
- Missing color/output/depth/motion handles.
- Vulkan command_buffer NULL or layout UNDEFINED.
- Ghosting: reset_history on hard cuts; check MV current→previous and +Y down jitter.
- Wrong output size: QUERY_STATUS and confirm scaler_factor.
- Compile error on enable: pass in2[2], not &in_frame.
- GEN handle recreate: in_frame[0].width/height must stay at render input size.

5. Unity / Unreal

- DllNotFoundException: place the native plugin under Assets/Plugins/Android/arm64-v8a/.
- UE missing library: lib/android and lib/ios libmagic_sr.a (the v2.1.0 enable archive no longer exists).
- Plugin Enable cannot feed graphic extras; use Native session API for graphic.

6. Diagnostic checklist

- mc_nscaler_version should report v3.0.0. Copy the string before FreeLibrary/dlclose.

- Record backend, input type, alg_mode, scale, width, height, model path.
- Record API return values and output_status_params_t.error_code.
- For graphic, confirm temporal_status_valid after a successful process.